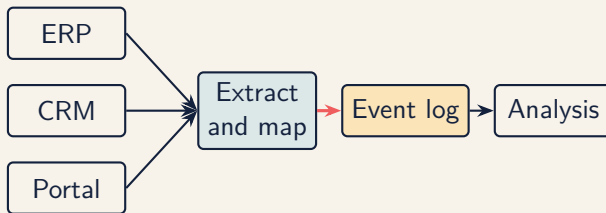


03

Event-Log Engineering and Data Quality

An event log is a designed analytical artifact, not a raw database export.

From source systems to an event log



Mapping decisions include:

- ▶ Which database change counts as a business event?
- ▶ How are records correlated into cases?
- ▶ Which timestamps and time zones are authoritative?
- ▶ Which low-level events should be abstracted?

Six common data-quality traps

1. Missing events
2. Duplicate events
3. Incorrect ordering
4. Mixed granularity
5. Unstable case identifiers
6. Selection and extraction bias

Diagnostic habit

For every surprising pattern, ask: *Is this process behavior, logging behavior, or extraction behavior?*

The case-notion test

A useful case notion should satisfy three questions:

1. Do the grouped events belong to the same process instance?
2. Does the resulting sequence have a meaningful beginning and end?
3. Does it support the decision the analysis is meant to inform?

Order as case

Good for customer lead time and order completion.

Item as case

Better for partial fulfillment and item-level handling.

A compact quality report

Check	Evidence	Decision / limitation
Case completeness	8.4% lack an end event	Exclude open cases from cycle-time KPI
Timestamp order	23 negative intervals	Correct known clock offset; flag remainder
Duplicates	1.2% exact duplicates	Remove with documented key
Activity meaning	"Update" has 4 sources	Split by business semantics
Coverage	Data begins Jan 2025	No claims about earlier process versions

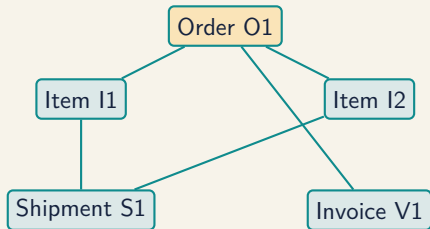
Key idea Documenting exclusions is part of the result, not administrative overhead.

Profile before you transform

Level	Profile	Decision it informs
Event	Missingness, duplicates, timestamps	Repair, exclude, or reinterpret
Case	Length, duration, completeness	Boundaries and open-case policy
Activity	Frequency, lifecycle, resources	Abstraction and filtering
Variant	Coverage, rarity, entropy	Noise strategy and segmentation
Temporal	Volume, drift, seasonality	Time windows and evaluation split

Key idea Profiling is not merely descriptive. It determines which transformations, models, and evaluation designs are defensible.

Flattening is a modeling choice



Flatten by order

- ▶ Shared item events converge

Flatten by item

- ▶ Shared order events duplicate

Keep objects

- ▶ Relationships remain explicit

Discuss Which case notion preserves the behavior needed by your question, and which frequencies does it distort?

04

Exploratory Process Analysis

Build a map of the evidence before building a process model.

Explore at four levels

Log

Cases, events, date range, coverage

Case

Duration, event count, outcomes

Activity

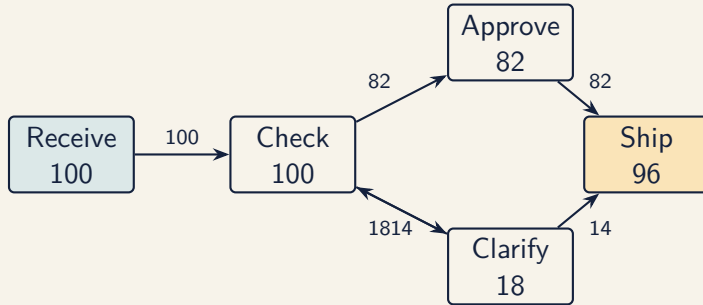
Frequency, resources, timing

Variant

Paths, prevalence, performance

Discuss Which level would reveal rework? Which would reveal a seasonal extraction gap?

A directly-follows graph



A DFG summarizes observed adjacency:

- ▶ Nodes are activities; edges mean “directly followed by”
- ▶ Counts can represent frequency or performance
- ▶ Loops, choices, and concurrency require careful interpretation

Variants: useful, but easy to misuse

Rank	Variant	Cases	Median time
1	Receive, Check, Approve, Ship	68%	2.1 d
2	Receive, Check, Clarify, Check, Ship	14%	5.8 d
3	Receive, Check, Reject	8%	1.4 d
4+	17 other variants	10%	8.2 d

Variants reveal

- ▶ Dominant routes
- ▶ Rework patterns
- ▶ Exceptional behavior

Variants can hide

- ▶ Concurrent behavior
- ▶ Attribute differences
- ▶ Important rare cases

A variant depends on its trace encoding

Encoding	Trace representation	Variants distinguish
Control-flow	Activity sequence	Routing differences
Lifecycle-aware	Activity + transition	Start, suspend, resume, complete
Time-aware	Activity + duration bucket	Behavioral and timing regimes
Payload-aware	Activity + selected attributes	Context or outcome cohorts
Learned	Sequence / payload embedding	Similarity beyond exact equality

Key idea Variant counts are not properties of the raw log alone. They are produced by an encoding, equivalence rule, and similarity threshold.

Profile the payload before using it

Event payload

Information carried by an event beyond case, activity, and timestamp.

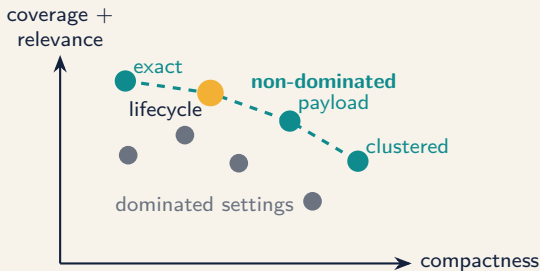
- ▶ Structured attributes and measurements
- ▶ Text, documents, images, audio, and sensor references
- ▶ Resource, cost, location, and product context

Discuss Use payload dimensions only when they serve the analytical question; otherwise they can fragment traces into meaningless one-case variants.

Profile before encoding

- ▶ Availability at event time
- ▶ Missingness and cardinality
- ▶ Drift, semantics, and cardinality
- ▶ Privacy, proxies, and future leakage

Pareto front for trace-variant detection



Candidate settings

Encoding, similarity threshold, rare-variant cutoff, and payload dimensions.

Objectives

Case coverage, compactness, decision relevance, and temporal stability.

Inspect representative traces before selecting a non-dominated setting.

A modeling-readiness profile

Before discovery or prediction, record:

- ▶ Case and trace-length distributions
- ▶ Variant coverage and long-tail behavior
- ▶ Class and outcome imbalance
- ▶ Missing and high-cardinality attributes
- ▶ Process and population drift
- ▶ Open versus completed cases
- ▶ Observation and label timestamps
- ▶ Potential future-information leakage

Deliverable

A one-page readiness report: usable population, known distortions, representation choice, split strategy, and excluded information.

PM4Py exploration

```
import pm4py

print(pm4py.get_start_activities(log))
print(pm4py.get_end_activities(log))

variants = pm4py.get_variants_as_tuples(log)
dfg, starts, ends = pm4py.discover_dfg(log)

filtered = pm4py.filter_case_performance(
    log,
    min_performance=5 * 24 * 60 * 60,
)
```

Interpretation task: compare common cases with slow cases. Which path differences are plausible explanations, and which need more evidence?